4CS017/UM1: Internet Software Architecture and Databases

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REFLECTIVE REPORT

Bolton OpenWeatherMap App

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Reflective Report:
Bolton Weather App Architecture

The architecture of the Bolton Weather App is simply based on a web-server model that retrieves live temperature data for Bolton the OpenWeatherMap API cloud service with the help of internet. The app contains three main components, i.e. **HTML and JavaScript** and runs directly on a web browser, **internet** for communication medium and **OpenWeatherMap** cloud server, which helps to process requests and provides real-time weather information.

Strength:

One of the main strengths of this architecture is its simplicity and lightweight. It does not require any complex set-up or server-side programming. The app only needs a web browser (which handles everything, from sending the request to displaying the temperature) and internet connection (which makes app lightweight, easy to use, and suitable to run on any device). Another strength is that it's modular and easy to update. If we needed extra weather features like wind speed, humidity, or weather icons can be added later without changing the whole structure. The reliance on HTTPS for communication proves that data transfer is secure. Also, the use of live data from OpenWeatherMap means the information is accurate and updated in real time.

Weaknesses:

However, there are several weaknesses as well. The application fully depends on internet connection and the OpenWeatherMap service. If the internet connection is slow and the API is not working properly, the application cannot function properly, also the API key is visible in JavaScript code, it means, it can be seen by anyone who views the page, which can be security risk. And another weakness is, the application doesn't work offline or store data, so it always needs to fetch new information.

Conclusion:

Overall, architecture of the Bolton Weather App is a good example of a simple and functional client- based design for small projects and educational purposes but would require optimization and include better security, caching, and offline support to make it more reliable.